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CROSS-DOMAIN SYNTHESIS

The Wall That May Not Be There

Flood engineers, reinsurers and ceramicists design against catastrophe with the same equation, and disagree only about the sign of one number.

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On 29 June 2021 the village of Lytton, in the dry canyon country of interior British Columbia, reached 49.6 degrees Celsius. The Canadian national record before that week had stood since July 1937 at 45.0 degrees, set on the Saskatchewan prairie. Lytton did not edge past it. Lytton beat it by 4.6 degrees, on the third consecutive day of breaking it, and burned down the next day.

What makes the number unsettling is not its size but its position. Statisticians who fit distributions to annual maximum temperatures for a living had fitted one to the Pacific Northwest, using seventy years of observations and allowing the distribution to drift upward with global mean temperature. The June 2021 reading did not sit in the far tail of that fitted curve. It sat outside it altogether, beyond the largest value the fitted distribution assigned any probability at all [1]. The model had not merely been surprised. It had contained a wall, and the weather had walked through the place where the wall was supposed to be.

That wall is a single parameter, and its sign is the subject of this piece. The same parameter, in a different profession, decides how high to build a sea dike; in a third, what a reinsurance layer is worth; in a fourth, how much strength a ceramic component loses when you scale it up. These trades do not read each other's journals. They are nevertheless solving the same equation, and almost all of their practical disagreements reduce to what they believe about one number.

Only Three Things a Maximum Can Do

The central theorem of the subject was proved in Cambridge in 1928 by Ronald Fisher and Leonard Tippett, and given its modern necessary-and-sufficient form by Boris Gnedenko in 1943 [2], [3]. It concerns the largest of n independent draws from some distribution. Nobody knows what that distribution is — that is the point — and for almost any interesting question nobody ever will.

The theorem says this. Suppose you take the maximum of n draws, recentre and rescale it so that it settles down as n grows, and suppose the result converges to something non-degenerate at all. Then that something must be one of exactly three distributions. Not approximately three. Three.

The reasoning is a stability argument of striking economy. The maximum of $2n$ draws is the maximum of two independent maxima of n draws each. So whatever limiting law G the maximum obeys, G^2 must be the same law with different centring and scale; the same for G^3 , and for every power. Very few functions survive that constraint, and solving for them yields the three families. Everything else about the parent distribution — its mean, its skewness, whether it is a sum of a thousand small effects or the product of a dozen large ones — washes out.

Anthony Jenkinson noticed in 1955 that the three families are one family wearing three hats, and wrote them as a single expression [4], now universal as the generalised extreme value distribution:

$$G(z) = \exp\left\{-\left[1 + \xi\left(\frac{z - \mu}{\sigma}\right)\right]^{-1/\xi}\right\}$$

with μ a location, $\sigma > 0$ a scale, and ξ the shape. The case $\xi = 0$ is read as the limit, giving the double-exponential Gumbel law. What was three qualitatively different answers becomes one continuous dial [5].

And the dial is not a refinement. It decides the question everyone actually cares about. When $\xi < 0$ the distribution has a finite upper endpoint,

$$z_{\max} = \mu - \frac{\sigma}{\xi}$$

above which nothing can ever be observed, no matter how long you wait. When $\xi > 0$ the tail is a power law with index $1/\xi$, unbounded, and heavy enough that high moments cease to exist. When $\xi = 0$ the tail is exponential: unbounded, but only barely, and with every moment finite. One number decides whether the worst case is a wall, a cliff or a slope.

Shape	Family	Tail behaviour	Typically fitted to
$\xi < 0$	Weibull	bounded above	material strength, annual maximum temperature
$\xi = 0$	Gumbel	exponential	storm surge, classical flood design
$\xi > 0$	Fréchet	power law	fire and liability losses, extreme rainfall

The names carry an irony worth noting. Emil Julius Gumbel, whose name sits on the middle row and who did more than anyone to turn this theorem into an engineering tool, first became known for a different kind of counting. In 1922 he published a tally of the political murders of the early Weimar Republic, finding that the overwhelming majority were committed from the right and that almost none of those were punished [6]. It cost him his chair at Heidelberg in 1932, and then his citizenship. He arrived at flood statistics as an exile, and the textbook that made the field practical appeared in New York in 1958 [7].

What A Return Period Actually Says

Engineering does not speak in shape parameters. It speaks in return periods: the hundred-year flood, the ten-thousand-year sea level, the fifty-year wind. A return period of T years names the level exceeded by one annual maximum in T , and inverting the expression above gives the level directly. Writing $y_T = -\ln(1 - 1/T)$ for the annual exceedance rate, the level is

$$z_T = \mu + \frac{\sigma}{\xi} (y_T^{-\xi} - 1), \quad z_T = \mu - \sigma \ln y_T \text{ when } \xi = 0$$

Two things about that phrase deserve more suspicion than they get. The first is elementary and still routinely misread. A one-in-hundred-year flood is not a flood that arrives every century. If exceedances are independent from year to year, the chance of at least one in a hundred years is $1 - (1 - 1/100)^{100}$, which is 0.6340 — and as T grows this converges to $1 - 1/e$, or 0.6321. Over a thirty-year mortgage the same flood has probability 0.2603. Gumbel was making this point to hydrologists in 1941 [8], and it still needs making.

BEFORE READING ON

A coastal authority has a century of annual maximum sea levels and must design for the level exceeded once in ten thousand years. How much of that answer is coming from the data?

The second is the real problem, and the figure below is the whole of it. Take three generalised extreme value distributions with identical location $\mu = 3.00$ metres and identical scale $\sigma = 0.30$ metres, differing only in shape: $\xi = -0.15$, $\xi = 0$ and $\xi = +0.15$. Over the range a century of annual records actually constrains, they are nearly the same distribution. At a ten-year return period their levels are 3.57, 3.68 and 3.80 metres — a spread of 23 centimetres, comfortably inside the sampling noise of a hundred observations. No goodness-of-fit test on such a record will confidently separate them.

Now extrapolate. At ten thousand years the three give 4.50, 5.76 and 8.96 metres. The spread is 4.46 metres: nineteen times what it was, and the difference between a dike and a decision to move the city. The $\xi = -0.15$ curve, moreover, is bounded by $\mu - \sigma/\xi = 5.00$ metres exactly. Not 5.00 metres at ten thousand years — 5.00 metres at any return period whatsoever, a level that a million years of storms cannot reach.

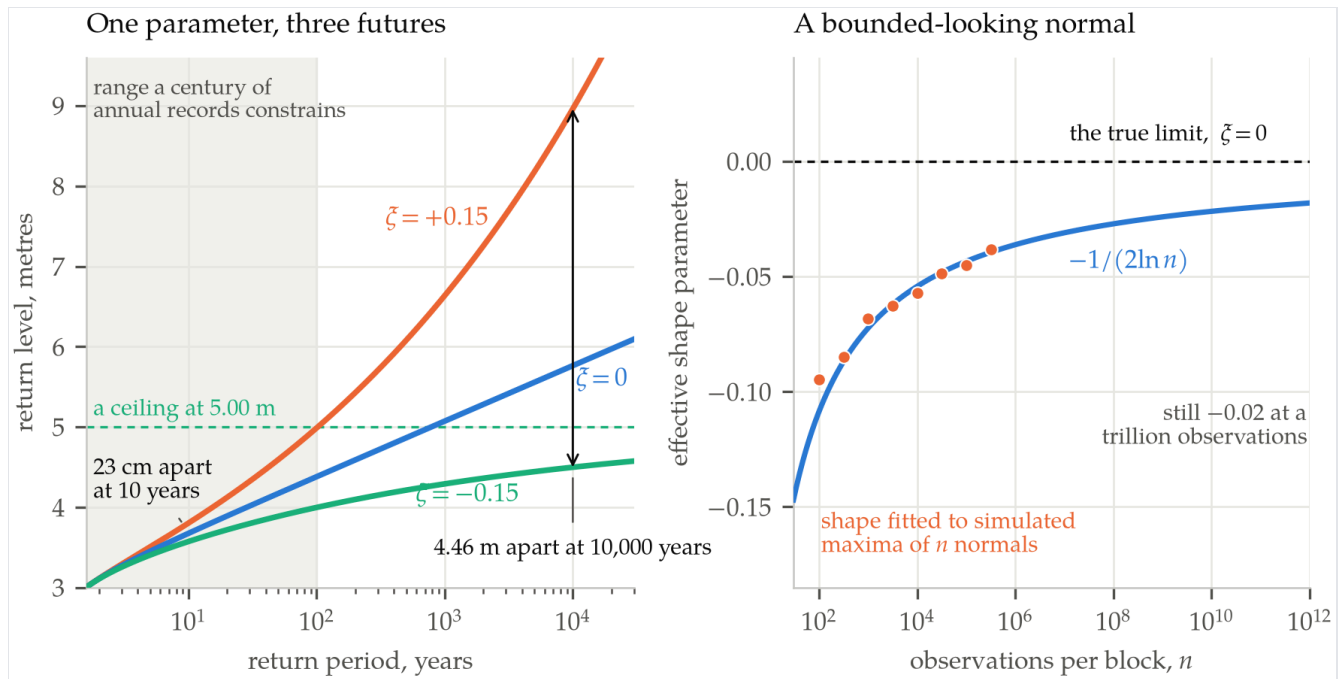


FIGURE Left: three GEV distributions with the same location and scale, differing only in shape, fanning out beyond the range the data constrain. Right: the shape parameter fitted to simulated maxima of n standard normal variables, against the penultimate value $-1/(2 \ln n)$.

The structure underneath is simple enough to state in words. Each factor of ten in rarity adds a fixed increment $\sigma \ln 10 = 0.691$ metres when $\xi = 0$; above zero, each factor of ten multiplies the excess over μ by 10^ξ asymptotically; below zero the increments shrink geometrically and sum to a finite total. The answer to the call-out is therefore uncomfortable. The data supply μ and σ with reasonable precision. They barely constrain ξ , and ξ is what the extrapolation runs on.

A Dike, An Economist And A Straight Line

The Dutch confronted this earlier and more seriously than anyone, because they had to. In 1939 Pieter Wemelsfelder, an engineer at Rijkswaterstaat, plotted half a century of high waters at Hoek van Holland on logarithmic paper and found that the exceedance frequencies fell close to a straight line [9]. He drew the correct and radical conclusion: design should be against a stated probability rather than against the highest level anyone happened to have seen, and on that basis many Dutch dikes were too low.

The argument was not acted upon. On the night of 31 January 1953 a North Sea surge overtopped and breached the defences of Zeeland and South Holland and killed 1,836 people. The Delta Committee convened afterwards did something no flood authority had done before: it asked a mathematician what the right probability was.

David van Dantzig's answer appeared in *Econometrica* in 1956 [10]. He wrote the present value of expected flood losses as damage multiplied by an exceedance probability that falls exponentially in dike height, added the cost of construction, and minimised the sum. It is a clean piece of operations research

— the marginal metre of dike is built when it costs less than the discounted risk it removes — and its descendants still govern Dutch investment decisions.

What happened next is the part worth keeping. His calculation for the heart of Holland pointed to an exceedance probability of one in 125,000 per year and a design level at Hoek van Holland of about 6 metres above the Amsterdam datum — a rise of 215 centimetres on the 3.85 metres then in place. The standard written into Dutch law in 1958 was one in 10,000 per year, twelve and a half times less demanding, and the level chosen was 5.00 metres, a rise of 115 centimetres. Just over half the height the economics recommended, and the economics was acknowledged at the time as one argument among several rather than the decisive one [11].

It is tempting to read that as politics overruling arithmetic, and that is partly what it was. But notice what the arithmetic rested on. Wemelsfelder's straight line on log paper is precisely the assumption that $\xi = 0$ — an exponential tail, the knife edge between a bounded world and a power-law one. It was adopted because the plotted points looked straight over the range that had been observed, which is exactly the range in which the figure above shows all three shapes looking alike. Later Dutch work took the shape parameter seriously as a quantity to be estimated rather than assumed, and Laurens de Haan's 1990 account of that effort remains the best description of what it is like to extrapolate a tail three orders of magnitude beyond your data with a country underneath it [12].

The Weakest Link And The Heaviest Loss

Two other professions found the same three families by walking in from opposite directions.

Waloddi Weibull was a Swedish engineer interested in why nominally identical specimens break at wildly different stresses. His answer, published in 1939 and restated for an American audience in 1951, was that a brittle body is a chain [13], [14]. It contains flaws of random severity, and it fails when the worst flaw in it fails. The strength of the body is therefore the *minimum* over its flaws — and a minimum is a maximum upside down, so the same theorem applies. The distribution that bears his name is what you get.

The consequence engineers care about follows in two lines. If a body of volume V survives stress σ with probability $\exp[-V(\sigma/\sigma_0)^m]$, then setting the survival probabilities of two sizes equal gives

$$\frac{\sigma_2}{\sigma_1} = \left(\frac{V_1}{V_2} \right)^{1/m}$$

Big things are weaker than small things made of the same material, because they contain more chances to be unlucky, and the exponent is the reciprocal of the Weibull modulus m . For ordinary glass, with m around 5, four decades of volume cost a factor of 6.31 in strength. For a well-made alumina at $m = 10$ the same scaling costs 2.51, and at $m = 25$ only 1.45. The modulus is effectively a quality metric: a

material with a high m is one whose worst flaw is predictable. Weibull's 1951 paper, it is worth recording, was received with open scepticism, and he conceded that he had no theoretical derivation and doubted one was available.

Insurance came in from the other side and found the third family. The relevant machinery is not block maxima but exceedances: given that a loss is larger than some threshold u , how much larger? James Pickands in 1975, and August Balkema and Laurens de Haan in 1974, established that the answer converges to the generalised Pareto distribution, with the same shape parameter ξ as the corresponding extreme value limit [15], [16]. Bruce Hill's estimator of that parameter, from the same year, remains the standard first look at a fat tail [17].

Here the sign of ξ is not about a ceiling; it is about whether the ordinary apparatus of insurance works at all. Under the generalised Pareto the mean loss is finite only when $\xi < 1$, and the variance only when ξ is below one half. Above that second threshold the law of large numbers stops being useful: pooling more policies no longer makes the average predictable, because the average is dominated by whichever single claim happens to be largest.

Which brings us to the best-studied loss dataset in the literature. Alexander McNeil's 1997 study of large Danish fire insurance claims — 2,156 losses above one million kroner — fitted a generalised Pareto above a threshold of 20 million and obtained $\xi = 0.684$, a tail index of $1/0.684 = 1.46$ [18]. Analyses of the same losses using Hill's estimator at a threshold of 10 put the index nearer 2.01, or $\xi \approx 0.50$; Sidney Resnick's companion discussion, printed immediately after McNeil's paper in the same issue, is largely about how unstable that choice of threshold makes the answer [19].

WORTH PAUSING ON

Careful analyses of one loss dataset, differing mainly in where they cut the threshold, straddle the exact value at which the variance of a fire loss becomes infinite.

That is not a failure of either analysis. It is what tail estimation looks like: the answer depends on where you cut the threshold, and there is no consistent estimator that avoids the choice. The practical reading is that Danish fire losses sit close enough to $\xi = 1/2$ that a reinsurer cannot know from the data whether the variance of its own portfolio exists — and a difference of 0.18 in ξ corresponds, deep in the tail, to a 1-in-1,000 loss that is 3.2 rather than 4.8 times the 1-in-100 loss, which is precisely the number a treaty is priced on [20].

The Fine Print In The Theorem

The theorem promises that the maximum converges to one of three laws. It says nothing whatever about how fast.

For the normal distribution, the news is bad in an interesting way. Normal maxima do converge, to the Gumbel law with $\xi = 0$. But Fisher and Tippet noticed in their original 1928 paper that the approach is so slow as to be useless, and that finite samples are better fitted by a *bounded* distribution with $\xi < 0$ [2]. The modern statement, sharpened by Joan Cohen in 1982, is that the best shape parameter for the maximum of n standard normals is approximately $-1/(2 \ln n)$ [21]: negative at every finite n , and creeping to zero at the speed of one over a logarithm.

The right-hand panel of the figure is that claim checked by simulation. Sixty thousand maxima were drawn at each of eight block sizes from a hundred to three hundred thousand, and a generalised extreme value distribution fitted to each set. The fitted shapes run from -0.095 to -0.038 and track $-1/(2 \ln n)$ throughout. The most unbounded distribution in statistics looks bounded at every sample size anyone will ever collect. Pushed the other way, the error in the limiting Gumbel approximation is still 3.1 per cent at a million observations per block and 2.0 per cent at a trillion; it takes n of order 10^{100} to get it under half a per cent.

So a fitted ξ of -0.05 is entirely consistent with a parent distribution that has no upper bound at all, and the apparent wall is an artefact of the sample size. It is also entirely consistent with a parent distribution that does have one. The data cannot distinguish these, and no amount of care with the fitting procedure will make them distinguishable, because the difference lives in a region where there are no observations by construction.

There is a second piece of fine print, and 2021 is where it bit. The theorem assumes the draws are independent and identically distributed. Temperature extremes under a warming climate are not identically distributed; they are drawn from a distribution whose location is moving. The Pacific Northwest analysis handled that in the standard way, letting μ shift with global mean temperature while holding σ and ξ fixed [1]. That is a reasonable assumption and it is still an assumption, and Lytton's 49.6 degrees fell outside the resulting curve's support. The team's own reading was that the event was made at least 150 times more likely by human-caused warming and that something about the fitted structure was wrong — either the shape, or the fixed scale, or the premise that a moving location is enough.

The honest summary of the whole business is this. Extreme value theory is a genuine and beautiful reduction: it tells you that the shape of catastrophe in your field is one of three things, and it tells you this without requiring you to know anything about the mechanism. What it does not do is tell you which of the three, and that is the only part anyone needs. A dike, a treaty and a turbine blade are all bets on the sign of ξ , placed by people who mostly do not realise they are betting on the same number, using data that by definition contains almost no information about it.

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